

WEEK 13

From a variant to an evidence-based sentence

How do we report uncertainty without turning it into a diagnosis?

By the end, you should be able to...

Distinguish a variant, a predicted consequence and a clinical classification.

Check reference, transcript, condition, date and review status.

Write a non-diagnostic interpretation that preserves uncertainty.

Reconnect with the biology

A sequence difference is a variant. Its effect depends on location, transcript, mechanism and biological context.

Think silently -> explain to a neighbour -> check the measurement.

One change, several descriptions

01

A variant can be described at genomic, transcript or protein level.

>

02

Different transcript choices can change a predicted consequence.

>

03

Record the reference accession and version before comparing labels.

Consequence is not clinical classification

Predicted consequence

Clinical significance

Missense means an amino-acid substitution in a specified coding context.

A stop-gain or splice prediction does not automatically establish pathogenicity.

Gene-disease mechanism and other evidence matter.

Five categories preserve uncertainty

Benign

Likely benign

Uncertain

Likely pathogenic

Pathogenic

Benign; likely benign; uncertain significance; likely pathogenic; pathogenic.

These categories were established for Mendelian germline variant interpretation.

Somatic treatment claims require their own framework and clinical context.

Evidence comes from different sources

01

Population frequency, segregation and functional studies can contribute.

>

02

Computational predictors provide another type of evidence.

>

03

Do not count correlated predictors as independent proof.

ClinVar records assertions and review status

Example status	What it describes
Expert panel	Level of review
Multiple submitters, no conflicts	Concordant assertions with criteria
Conflicting classifications	Disagreement to examine

Check the exact variant, condition, submission dates and supporting evidence.

Conflicts or older assertions may require investigation.

Review stars describe review status, not a pathogenicity probability.

A VUS is not a treatment instruction

Uncertain significance means available evidence is insufficient or conflicting.

Do not use a VUS alone to establish a diagnosis or change treatment.

Escalate real clinical questions to qualified clinical professionals.

What a useful sentence contains

01

Identity: the exact variant and reference context.

>

02

Evidence: what supports the current interpretation and what is missing.

>

03

Scope: an educational interpretation, not clinical advice.

Our cases are deliberately fictional

EDU-V1, EDU-V2 and EDU-V3 are synthetic evidence cards.

They have no real patient identifiers or claimed ClinVar accessions.

The goal is to audit reasoning, not independently certify a clinical classification.

Worked example: a cautious report

EDU-V1 is a rare fictional missense variant with only two correlated computational predictions and no functional or segregation evidence.

Predict the result and one limitation before calculating.

Work through the reasoning

Rarity and predicted damage do not alone establish disease causality.

The evidence is insufficient for a confident pathogenic interpretation.

Report the uncertainty and request relevant independent evidence; do not diagnose.

Pause: make a defensible claim

A variant is rare and a predictor calls it damaging. Does this establish pathogenicity? Name one independent type of evidence to investigate.

First answer alone. Then compare the evidence, not just the conclusion.

A defensible response

No. Examples include appropriate functional evidence, segregation with the relevant phenotype, or well-curated condition-specific evidence evaluated in context.

Today's practical

W13 studio: inspect three synthetic variant cards and write three short evidence-bounded statements. No patient data or clinical recommendation is required.

Minutes	Action
0-10	Predict + set up
10-30	Run + inspect + check
30-40	Explain + compare
40-45	Save + submit / exit

Five terms to use precisely

- 1 variant
- 2 transcript
- 3 missense
- 4 classification
- 5 VUS

Exit ticket

What did we actually measure or calculate?

What is one assumption or limitation of that result?

What check would you perform before sharing it?

Go further only after the core is clear

Optional: explore current gene-specific ClinGen specifications as a supervised follow-on topic.

Not required for the core grade. No advanced model training is expected.

Read the evidence, not only the slide

S03 SAM/BAM, VCF and BED specifications, maintained by the HTS community.

S18 NCBI ClinVar: review status and representation of classifications.

S19 Richards et al. (2015). ACMG/AMP standards and guidelines for interpretation of sequence variants.